

Lab 06: Develop Your Google Cloud Network - Challenge Lab

Challenge Lab | Develop Your Google Cloud Network | GSP321

 **This is a Challenge Lab. No step-by-step instructions provided.**
Skills from Labs 01-05 were applied independently to complete all tasks.

Overview

Helped the Griffin team set up their development environment on Google Cloud. Created development and production VPCs, a bastion host, Cloud SQL instance, Kubernetes cluster, and deployed a WordPress application with monitoring enabled.

Tasks Completed

☒ Task 1 - Create development VPC manually ☒ Pass

☒ Task 2 - Create production VPC manually ☒ Pass

☒ Task 3 - Create bastion host ☒ Pass

☒ Task 4 - Create and configure Cloud SQL Instance ☒ Pass

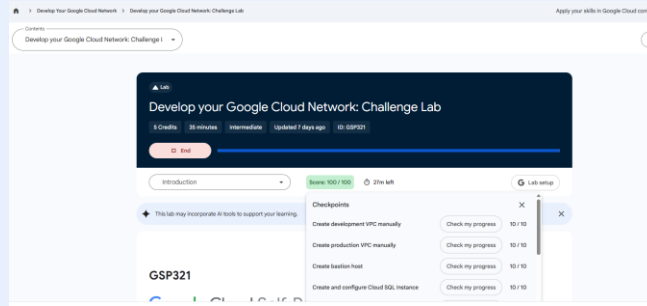
☒ Task 5 - Create Kubernetes cluster ☒ Pass

☒ Task 6 - Prepare the Kubernetes cluster ☒ Pass

☒ Task 7 - Create a WordPress deployment ☒ Pass

☒ Task 8 - Enable monitoring ☒ Pass

☒ Task 9 - Provide access for an additional engineer ☒ Pass



Task 1 and 2 - Create Dev and Prod VPCs

```
gcloud compute networks create griffin-dev-vpc --subnet-mode=custom
gcloud compute networks subnets create griffin-dev-wp --network=griffin-dev-vpc --region=[REGION] --range=192.168.16.0/20
gcloud compute networks subnets create griffin-dev-mgmt --network=griffin-dev-vpc --region=[REGION] --range=192.168.32.0/20
```

VPC networks

Create VPC network

Refresh

Learn

Networks in current project

Subnets in current project

VPC networks

Manage flow logs

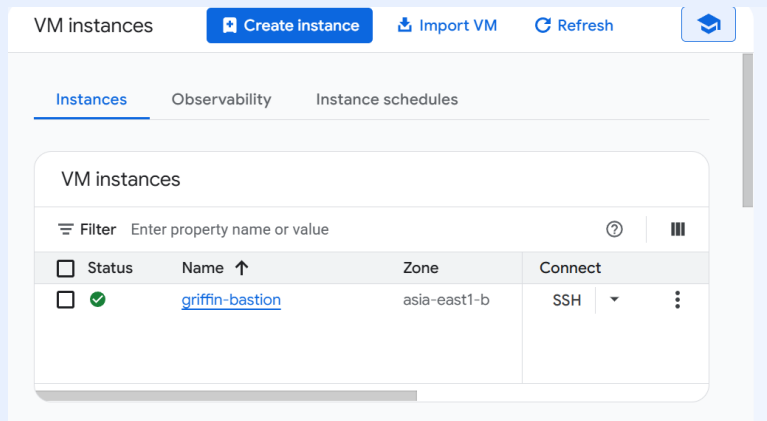
Filter

Enter property name or value

☐	Name ↑	Subnets	MTU	Mode	IPv6 ULA range	Gateways	Flow log configs
☐	griffin-dev-vpc	2	1460	Custom			
☐	griffin-prod-vpc	2	1460	Custom			

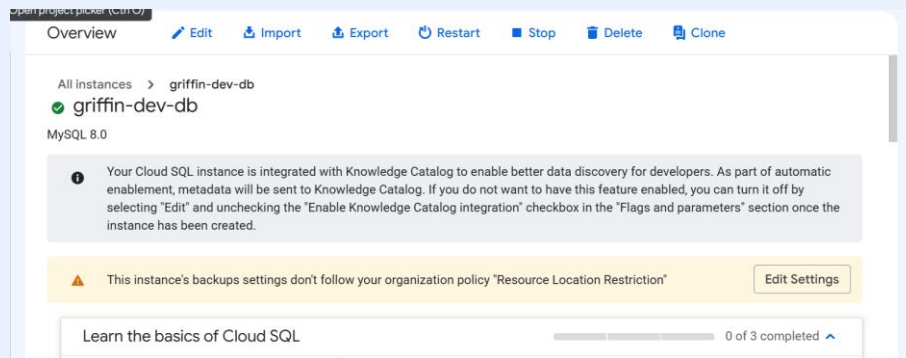
Task 3 - Create Bastion Host

Created bastion host with two NICs connected to griffin-dev-mgmt and griffin-prod-mgmt. Created firewall rules to allow SSH.



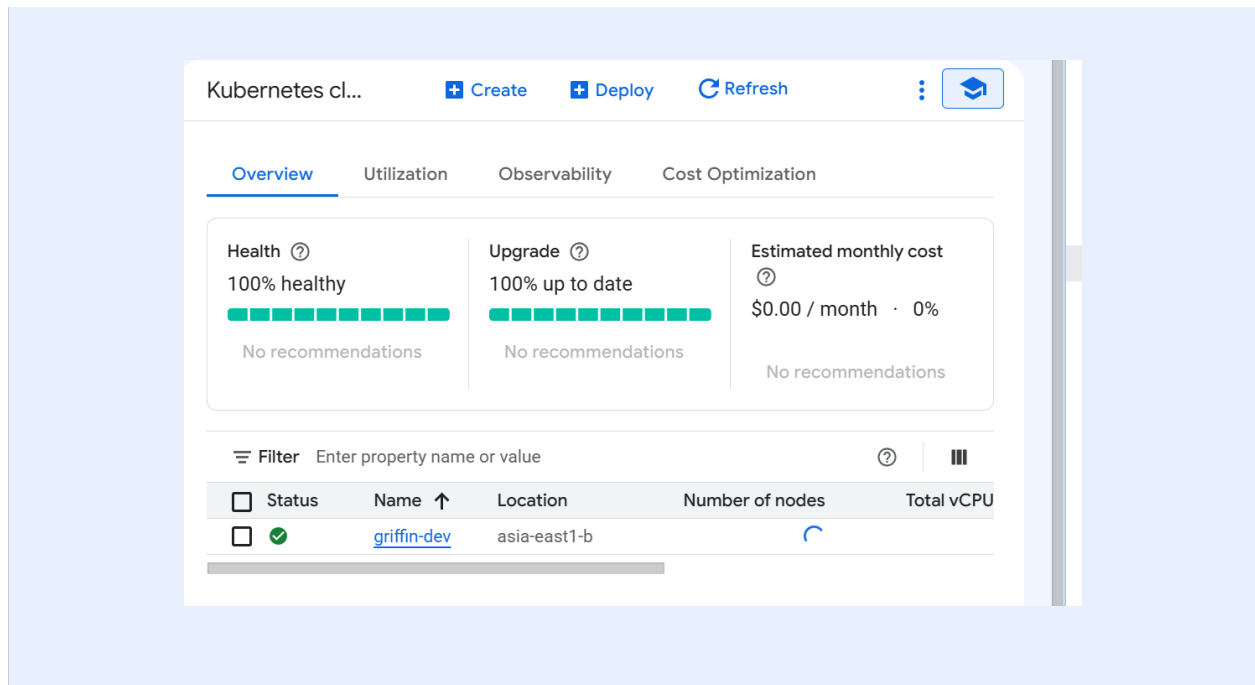
Task 4 - Cloud SQL Instance

```
CREATE DATABASE wordpress;
CREATE USER "wp_user"@"%" IDENTIFIED BY "stormwind_rules";
GRANT ALL PRIVILEGES ON wordpress.* TO "wp_user"@"%";
FLUSH PRIVILEGES;
```



Task 5 - Create Kubernetes Cluster

```
gcloud container clusters create griffin-dev --num-nodes=2 --machine-type=e2-
standard-4 --network=griffin-dev-vpc --subnetwork=griffin-dev-wp --zone=[ZONE]
```

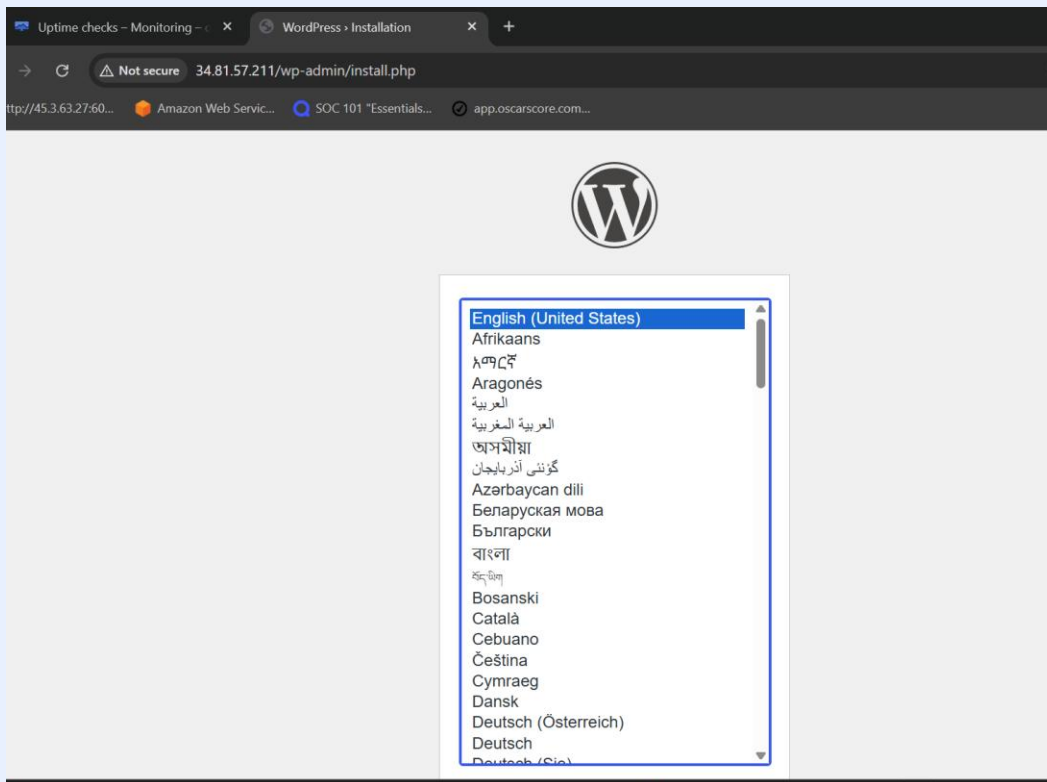


Task 6 - Prepare Kubernetes Cluster

```
gsutil cp -r gs://spls/gsp321/wp-k8s .  
gcloud iam service-accounts keys create key.json --iam-account=cloud-sql-  
proxy@$GOOGLE_CLOUD_PROJECT.iam.gserviceaccount.com  
kubectl create secret generic cloudsql-instance-credentials --from-file key.json
```

Task 7 - WordPress Deployment

Replaced YOUR_SQL_INSTANCE with griffin-dev-db instance connection name, applied wp-deployment.yaml and wp-service.yaml.



Task 8 and 9 - Monitoring and Access

Created uptime check for WordPress site. Granted second user account Editor role on the project.

```
role: roles/compute.serviceAgent
- members:
  - serviceAccount:service-236717431464@gcp-sa-gkenode.iam.gserviceaccount.com
  role: roles/container.defaultNodeServiceAgent
- members:
  - serviceAccount:service-236717431464@container-engine-robot.iam.gserviceaccount.com
  role: roles/container.serviceAgent
- members:
  - serviceAccount:236717431464-compute@developer.gserviceaccount.com
  - serviceAccount:236717431464@cloudservices.gserviceaccount.com
  - user:student-01-a51b89fe7f81@qwiklabs.net
  role: roles/editor
- members:
  - serviceAccount:service-236717431464@gcp-sa-networkconnectivity.iam.gserviceaccount.com
  role: roles/networkconnectivity.serviceAgent
- members:
  - serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
  - serviceAccount:qwiklabs-gcp-01-7cbbf7eaed5b@qwiklabs-gcp-01-7cbbf7eaed5b.iam.gserviceaccount.com
  - user:student-01-alb314077f9b@qwiklabs.net
  role: roles/owner
- members:
  - user:student-01-alb314077f9b@qwiklabs.net
  role: roles/resourcemanager.projectIamAdmin
- members:
  - serviceAccount:qwiklabs-gcp-01-7cbbf7eaed5b@qwiklabs-gcp-01-7cbbf7eaed5b.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-01-alb314077f9b@qwiklabs.net
  - user:student-01-a51b89fe7f81@qwiklabs.net
  role: roles/viewer
etag: BwZT756wtis=
```

What I Learned

- Custom VPCs require manually creating subnets unlike auto mode networks
- A bastion host with multiple NICs bridges two isolated VPC networks securely
- Cloud SQL secrets must be configured before Kubernetes deployments can use them
- WordPress on Kubernetes requires both a deployment and a service YAML
- Uptime checks combined with IAM access management are core DevOps practices

Resources

GCP GKE Docs: <https://cloud.google.com/kubernetes-engine/docs> | Lab ID: GSP321